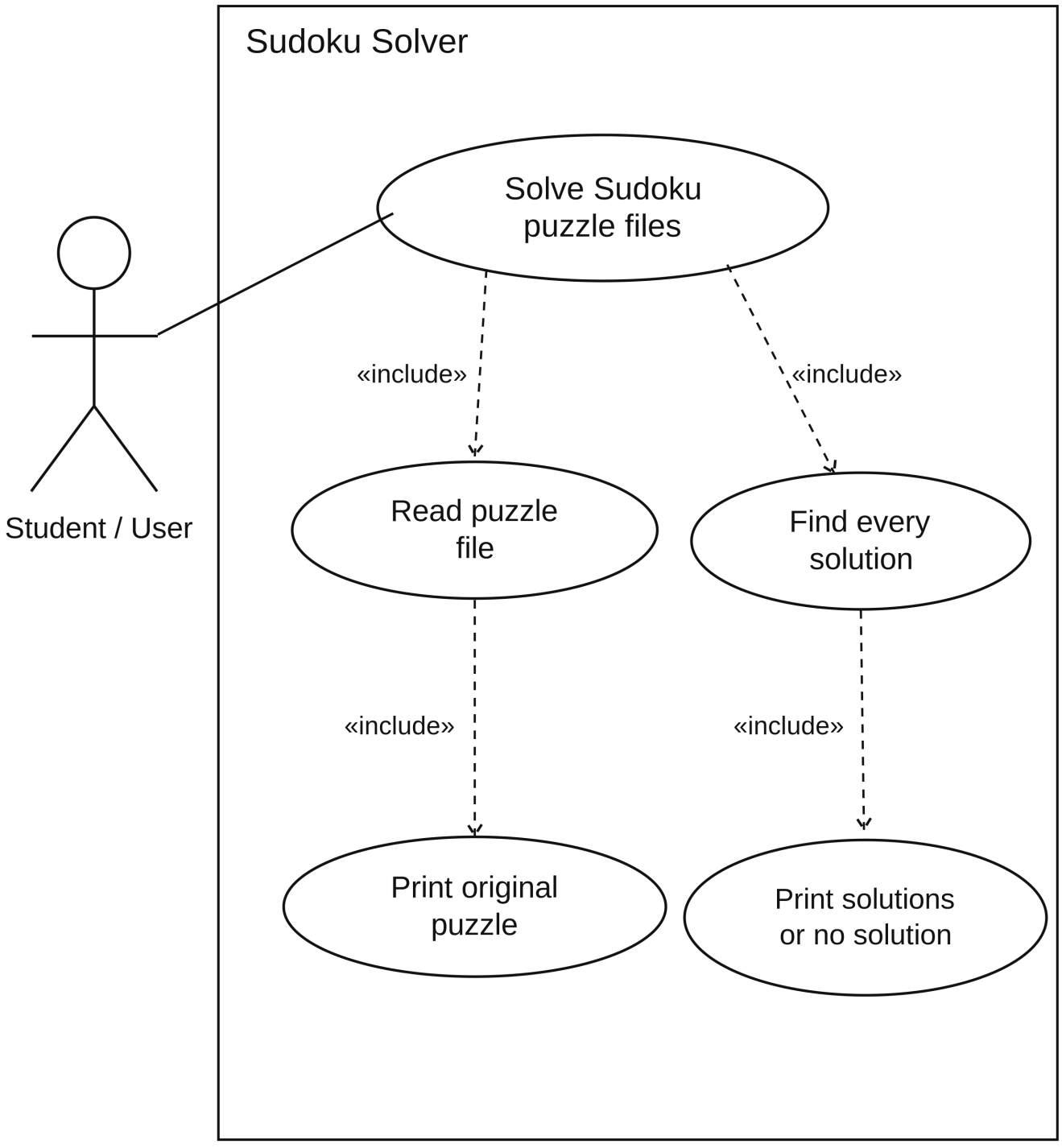


Assignment 4 USE CASES.

Use Case: Solve Sudoku Puzzles

The Actor: User

Trigger: The user runs Assignment4.exe.

Inputs: puzzle1.txt, puzzle2.txt, puzzle3.txt, puzzle4.txt, puzzle5.txt

Precondition: The puzzle files are in the same folder as the program.

Description:

The program reads all of the five Sudoku puzzle files and prints each original puzzle individually. If a puzzle has no solution, the program prints "No solution found". It uses recursive backtracking to find all of the possible solutions.

Main Steps:

1. The user runs “Assignment4.exe”
2. The program opens “puzzle1.txt”
3. It reads the 9x9 grid; the Empty cells are marked with underscores “_”
4. The program prints the file name and the original puzzle
5. It checks the starting numbers for any differences or conflicts
6. It finds an empty cell and tries all possible numbers from 1 to 9 recursively. The program backtracks and tries another number if the number provided doesn’t work
7. The program prints all possible valid solutions or "No solution found" if there are no valid solutions
8. It repeats the process recursively for the four files that remain.

Alternative Outcomes:

1. The program displays "Unable to read puzzle file" and moves to the next file, If the provided file is missing or invalid,
2. It prints "No solution found", if the puzzle cannot be solved.
3. If there is more than one solution, the program prints all of them individually

Postcondition:

The program attempted all five puzzle files and displayed all 5 results.